
Designing CRISPR Guide RNA

Gene of interest – PCSK9

ABOUT

PCSK9 (Proprotein Convertase Subtilisin/Kexin Type 9) is an enzyme encoded by the *PCSK9* gene in humans.

Location: chromosome 1p32.3

Size: 25 kilobases of genomic DNA, and comprises 12 exons that encode a 692-amino-acid protein.

Conservation across species: PCSK9 is highly conserved among mammals, including chimpanzee, monkey, camel, alpaca, rat, and mouse, with an approximate amino acid identity of 99%, 96%, 82%, 81%, 77%, and 77%, respectively.

Protein domains: The mature form of PCSK9 consists of a signal peptide (SP), a pro-segment (PRO), a catalytic domain, and a cysteine/histidine-rich domain (CHRD) in the C-terminal region. The main organ producing PCSK9 is the liver, followed by the kidneys and intestines.

How it works: The enzymatic activity of PCSK9 is mediated by its catalytic domain, which recognises specific structural features on the LDL receptor. Interestingly, the protein retains its receptor-degrading function even after its catalytic activity is disrupted by autocatalytic cleavage — meaning it is the *physical interaction* between PCSK9 and the LDL receptor, rather than ongoing proteolytic activity, that drives receptor degradation.

Why Knock Out PCSK9 using CRISPR?

PCSK9 promotes degradation of the low-density lipoprotein receptor (LDLR) and plays a central role in regulating plasma levels of LDL cholesterol, lipoprotein(a), and triglyceride-rich lipoproteins, thereby increasing the risk of cardiovascular disease.

Elevated PCSK9 levels impair LDL clearance by decreasing the availability of LDL receptors on hepatocytes, contributing to hypercholesterolemia and cardiovascular disease risk.

Meaning: High PCSK9 → High LDL-C

↑ Atherosclerosis risk

↑ Coronary artery disease

↑ Stroke, heart attack

↑ Peripheral artery disease

↑ Aortic aneurysm risk

Why CRISPR over drugs? While antibodies or siRNA against PCSK9 can significantly reduce LDL-C and cardiovascular events, they require regular, long-term administration. A CRISPR knockout, by contrast, is potentially a one-time permanent intervention. By precisely editing key genes such as PCSK9, CRISPR-Cas9 offers the possibility of long-term regulation of LDL cholesterol, which may reduce the risk of cardiovascular diseases.

❖ Steps for Designing a Guide RNA Against PCSK9

1. Get your gene:

Choose your target gene i.e. PCSK9

- Search NCBI Nucleotide
- Search PCSK9 Gene
- Download FASTA Format
- Save the file as “PCSK9.gene”

2. Visualization & Inspection using SnapGene Viewer:

Convert your raw FASTA Format into visual DNA map for proper understanding.

- File → open → “PCSK9.gene”

3. Designing Guides:

- Preferred tools – CHOPCHOP, CRISPOR, BENCHLING

- Choose your target region: Critically Exon1 and Exon2 of PCSK9 are strongly preferred but initial 500 sequences are generally enough to design a guide.
- Open CHOPCHOP
- Input: Paste FASTA Sequence → Select species(i.e. human) → Choose Cas9 / Cas12a / Cas12b.(design the guide using each Cas enzyme to select the best one)
- Configure: Guide length i.e. Cas9- 20 nt., Cas12a- 23 nt.

Extract the 20 nt spacer. The basic goal in sgRNA design is to select a 20 nt target sequence immediately upstream of a PAM site. The complementary 20 nt spacer RNA directs the Cas9 nuclease to the specific genomic location to be edited. The target sequence should be unique within the genome to avoid off-target effects.

- Run & Review: Download the Result table with hundred's of possible guide sequence.

4. Scoring & Evaluating the Guides:

- Identify PAM Sites: Cas9 – NGG, Cas12a – TTTN, Cas12b – TTN

The 3' end of the DNA target sequence must have a protospacer adjacent motif (PAM) sequence (5'-NGG-3') for SpCas9. Scan both strands of your chosen exon sequence for all NGG occurrences — each one defines a potential cut site.

- EFFICIENCY: Doench / CFD Score- >50% or above 70%
- OFF TARGETS: MMO- 0 or Total off targets under 10

Sequences fitting the PAM with fewer than three nucleotide mismatches across the genome are counted. Sequences with only one mismatch imply high off-target potential, while sequences of two or three mismatches should be limited. Mismatches closer to the PAM sequence carry higher weights.

- GC CONTENT: 40% – 60%
- SELF COMPLEMENTRITY – 0 or 1

The spacer sequence must not fold back on itself or disrupt the scaffold's stem-loops. Guide-internal base pairs interfere with DNA target binding.

Above evaluation can be done in Excel.

Note- For diagnostic applications, always validate across 2 tools before finalizing guides.

5. Copy the Guides from Excel and view it in SnapGene Viewer.

- Highlight the Spacer sequence and PAM using different colours.
- Repeat the same with all 3 Cas enzymes.

6. OUTPUT: Compile your output of top Guides into a structural report.